

n=89 Aug 57
McLan 61

Show Your Work

Exam I Sept 18th 2025 Chem 1127

Name (Full First & Last or -1 pt):

Key

4 pg Exam + 2 pg Equations/Periodic Table

Pts: ____ / 80

Page 1: ____ / 24

1. An unknown element is a mixture of isotopes ^{129}X and ^{132}X . The average atomic mass of X is 130.79 amu. What is the percent abundance of ^{132}X ? (3 pt)

40.3% or 40% as
answer (-1.5)

Decimal Abundance of isotope ^{129}X

$$r + p = 1 \rightarrow r = 1 - p$$
$$129 \times r + 132 \times p = 130.79$$
$$129(1 - p) + 132p = 130.79 \rightarrow 129 - 129p + 132p = 130.79$$
$$3p = 1.79 \rightarrow p = 1.79/3 = 0.597 \text{ or } 59.7\%$$

a. If there are only two isotopes of X, what must their % abundances add up to equal? (1 pt) 100%

2. For the $n = 3$ level/shell of an atom, clearly identify all possible values for ℓ , and for each ℓ clearly identify all possible values of m_ℓ . (2 pt)

1 mistake (-.5)
either let for wrong (-1)

$\ell = 0$
 $\downarrow$
 $m_\ell = 0$

$\ell = 1$
 $\downarrow$
 $m_\ell = -1, 0, 1$

$\ell = 2$ not clear (-.5)
 $\downarrow$
 $m_\ell = -2, -1, 0, 1, 2$

- a. For one of the values of ℓ , identify the letter associated with it, and how many orbitals are in this sublevel/subshell. (1 pt) letter of $\ell =$ # of orbitals =

orbitals
 $\ell = 0 \rightarrow s - 1$
 $\ell = 1 \rightarrow p - 3$
 $\ell = 2 \rightarrow d - 5$

3. For the three quantum numbers (n, ℓ, m_ℓ, m_s) match the symbol with the associated trait. (2 pt)

- a. Spin: m_s b. Direction/Orientation: m_ℓ c. Energy/Size: n d. Shape: ℓ

4. Which orbital has a dumbbell/hourglass shape (pictured here). Circle it. (1 pt)

- a. s b. f c. p d. d



5. An ion, $^{55}\text{Fe}^{2+}$, contains how many protons (p), neutrons (n), and electrons (e) and identify this as an anion or cation. (4 pt)

protons (p) 26 neutrons (n) 29 electrons (e) 24 anion or cation.

6. propane is (C_3H_8). Identify it as a(n) Compound. Circle the correct term. (2 pt)

- a. homogeneous mixture b. heterogeneous mixture c. element d. compound

7. Briefly define the term homogeneous mixture and provide an example of it. Your definition should incorporate the bolded prefix of the first word. (3 pt)

A mixture of different compounds and/or elements that has the same characteristics or properties throughout
Gatorade, Salt Water

8. Circle the property below that is a chemical property. (2 pt)

- a. Melting Point b. Density c. Toxicity d. Color

9. volume is an intensive or extensive property. Identify it and explain your identification. (1.5 pt)

Extends only to the sample at hand

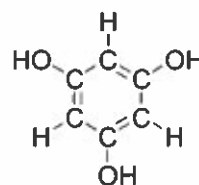
10. What happens to orbitals as n goes from $n = 1$ to $n = 2$. Define orbital and explain this change. (1.5 pt)

From Smaller to Larger
or
From Lower to Higher Energy
3D place where e^- likely/probably are.

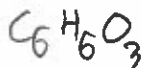
Show Your Work

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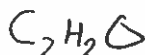
11. Given the structure shown here, clearly write its molecular and empirical formula. They may be the same thing. (3 pt)



Molecular Formula



Empirical Formula



12. Circle the below element(s) that is (are) diatomic in their elemental form. (2 pt)

1 wrong
2 wrong

Ca

Br

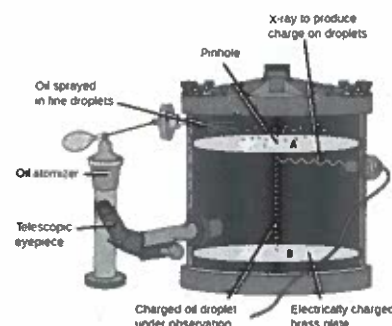
N

He

- a. For one of the diatomic elements, write the diatomic formula below. (1 pt)



13. In the given image of the oil drop experiment, identify what charge the top plate A has: negative, positive, or neutral. Circle it. Note the oil drops have a negative charge (due to the x-rays). (1 pt)



- a. Indicate what direction the oil drop travels if no electric field is applied between plate A & B. (1 pt)

Downward.

- b. This experiment identified the F.C. of the subatomic particle, the e^- . Circle the below terms to fill in the blanks. (2 pt)

Mass to charge ratio

Fundamental Charge

Electron

Proton

Nucleus

Neutron

14. Opposite charges Attract. Fill in the blank with a correct verb/word. (1 pt)

15. For the following blanks fill them in with proton (p), electron (e), or neutron (n). Note, one or more of the blanks should be left blank. (3 pt)

- a. e^- & $+$ are the lightest or of equally light mass.
b. p & $+$ is (are) positively charged.
c. p & n reside inside the nucleus of an atom.

All or nothing
each row

16. How many oxygen atoms are present in one molecule ("formula unit") of $Fe_2(Cr_2O_7)_3$? (1 pt)

21

- a. How many oxygen atoms would be present in 7 molecules of it? (1 pt)

$7 \times 21 = 147$

17. If the frequency of electromagnetic radiation increases, how does the wavelength & energy change? Do they stay constant, increase, or decrease? (2 pt)

wavelength = λ

energy = E

$E = h \cdot \nu$

Decrease

Increase

2

each 1pt

Show Your Work

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3 sig Figs = 0.5 pt

Key

18. Blue light has a wavelength of 425 nm, what is the frequency in Hz of this light? Provide your answer with the correct amount of significant figures. (3 pt)

$$\lambda = \frac{425 \text{ nm}}{1} \times \frac{10^{-9} \text{ m}}{1 \text{ nm}} = 425 \times 10^{-9} \text{ m}$$

$$c = v \times \lambda \rightarrow v = \frac{c}{\lambda} = \frac{3 \times 10^8 \text{ m/s}}{425 \times 10^{-9} \text{ m}}$$

$$v = 7.06 \times 10^{14} \text{ Hz}$$

- a. How can the units of Hz be rewritten in terms of seconds (s)? (1 pt)

$$1 \text{ Hz} = \frac{1}{\text{s}} = \text{s}^{-1}$$

- b. What is the energy of 1.3 moles of the above photons given the wavelength. (1.5 pt)

$$E = h \cdot v = 6.63 \times 10^{-34} \text{ J} \cdot 7.06 \times 10^{14} \text{ s}^{-1} = 4.68 \times 10^{-19} \text{ J/photon}$$

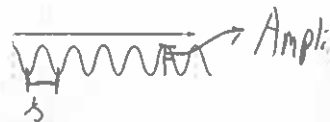
$$1.3 \text{ mol} \times \frac{6.02 \times 10^{23} \text{ photons}}{1 \text{ mol}} \times \frac{4.68 \times 10^{-19} \text{ J}}{1 \text{ photon}} = 3.66 \times 10^5 \text{ J}$$

- i. Is the energy equation on the equation sheet, you are using here, for 1.3 mol of photon, 1 mol photon or 1 photon? Circle the correct number. (0.5 pt)

19. An unknown element Y has the following isotopes: ^9Y (15.00% abundant, atomic mass = 9.0 amu) and ^{11}Y (85.00% abundant, atomic mass = 11.0 amu). What is the average atomic mass in amu of Y? (2 pt)

$$9 \times 0.15 + 11 \times 0.85 = 10.7 \text{ amu} = \text{AAM}$$

20. On the given diagram, identify wavelength and amplitude, and given that this shows how many waves there are in a 3 second interval, identify what the frequency is. (3 pt)



$$v = \frac{\text{Waves}}{\text{s}} = \frac{6 \text{ waves}}{3 \text{ s}} = \frac{2 \text{ waves}}{\text{s}}$$

21. For a hydrogen atom, calculate the wavelength of light that would be emitted for the orbital transition of $n(\text{initial}) = 4$ to $n(\text{final}) = 2$. Submit an answer to 3 significant figures. (3 pt)

$$\frac{1}{\lambda} = R_{\infty} \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$486 \text{ nm}$$

$$4.86 \times 10^{-7} \text{ m}$$

$$\lambda = \frac{1}{1.097 \times 10^7 \text{ m}^{-1} \left(\frac{1}{2^2} - \frac{1}{4^2} \right)} = \dots$$

22. In the photoelectric effect, if the amplitude (intensity) of light shone on a metal increases, what will happen? Circle it (2 pt)

a. Ejected electrons (e^-) will be faster

c. Ejected (e^-) will be slower

b. There will be more (e^-)

d. There will be fewer (e^-)

e. No change

23. The photoelectric effect above, is an example of the Law of the Conservation of energy. Fill in the blanks with the words below. (1 pt)

Generation Promulgation

Conservation

Distortion Theory

Hypothesis

Law

Amendment

24. Clearly identify how many significant figures are in the two values below. (2 pt)

a. 2.605×10^{-4}

b. 0.002500

4

4

25. Under standard conditions, what is the boiling point, in $^{\circ}\text{C}$, of water. (1 pt)

a. 100

b. 1

c. 0.0001

d. 0

Show Your Work

Key

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26. How many moles of ethanol are in 35.6 g of ethanol (C_2H_6O)? (2 pt)

$$\frac{35.6 \text{ g } C_2H_6O}{1} \times \frac{1 \text{ mol } C_2H_6O}{46 \text{ g } C_2H_6O} = 0.774 \text{ mol } C_2H_6O$$

a. The problem above requires you to identify the molar mass (MM) of ethanol. Do so and write it as a conversion factor with both units. If it is in the main part, that is fine. (2 pt)

$\frac{g}{mol} = 1 \text{ pt}$ Value $v/g = 1 \text{ pt}$

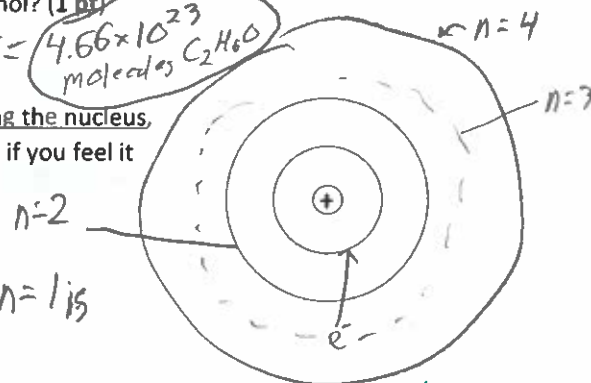
$$MM = 2 \times 12 + 6 \times 1 + 1 \times 16 = 46 \text{ g } C_2H_6O$$

$$\frac{46 \text{ g } C_2H_6O}{1 \text{ mol } C_2H_6O}$$

b. How many molecules of ethanol are present in 35.6 g of ethanol? (1 pt)

$$0.774 \text{ mol } C_2H_6O \times \frac{6.02 \times 10^{23} \text{ molecules}}{1 \text{ mol } C_2H_6O} = 4.66 \times 10^{23} \text{ molecules } C_2H_6O$$

27. Given this image of the Bohr model, with the innermost circle indicating the nucleus, identify the orbit corresponding to $n = 2$, you may explain your answer if you feel it necessary. (1 pt)



a. Is the above orbit the lowest energy orbit? Yes or No (1 pt)

$n=1$ is

b. Draw in the orbit for $n = 4$, or circle "not allowed". (1 pt)

c. An electron moving from $n = 3$ to $n = 2$ or to $n = 1$ involves more energy being released or absorbed. Circle the correct bolded/underlined term to make the statement true. (1 pt)

Full pt
Further Drop
means Larger
Energy Change

28. Orbitals hold a maximum of 2 electrons. Fill in the blanks with the correct number and subatomic particle. (1 pts)

29. An atom of a particular element has a volume of $6.1 \times 10^{-4} \text{ nm}^3$. What is this volume in cubic inches. (3 pt)

$$1 \text{ m} = 3.28 \text{ ft}$$

$$1 \text{ ft} = 12 \text{ in}$$

$$\frac{6.1 \times 10^{-4} \text{ nm}^3}{1} \cdot \frac{(10^{-9})^3 \text{ m}^3}{1^3 \text{ nm}^3} \cdot \left(\frac{3.28 \text{ ft}}{1 \text{ m}} \right)^3 \cdot \left(\frac{12 \text{ in}}{1 \text{ ft}} \right)^3 = 3.7 \times 10^{-26} \text{ in}^3$$

a. Write a conversion factor (fraction or ratio) for the above question. (1 pt)

$$\frac{1 \text{ m}}{3.28 \text{ ft}}$$

Flipped (-) $v n = 5$
no n^3 to

b. Identify one of the given numbers above as a measured number/value. (1 pt)

$$6.1 \times 10^{-4} \text{ nm}^3$$

c. Identify one of the given bolded units as a volume and one as a length/distance. (1 pt)

$\text{in}^3 \leftrightarrow \text{nm}^3$ m or in or ft

30. Is this a chemical reaction per Dalton's Atomic Theory? Why or why not? (2 pt)



Yes all atoms are conserve

31. What are columns and rows on the periodic table called? Clearly identify each from the words below. (2 pts)

Periods

Rows

Blocks

Groups

Columns

Domains

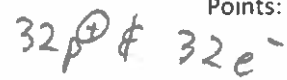
Just circled 4

Id'd but n/sd (+)

if $\uparrow$ R ... (-1.5) ... $\rightarrow$
Both right but

(-1.5)

1. Write the complete electron configuration for germanium (Ge). (2 pt)



1. Write

2. Write

$$[Ne] 3s^2 3p^1$$

Hund's rule
all unpaired in
same spins
boxes or a
Be sure

C_r

(1 _s)	(1 _s)	(1 _p)	(1 _p)	(1 _p)	(1 _p)	(1 _p)	(1 _p)	(1 _p)	(1 _p)	(1 _d)	(1 _d)	(1 _d)	(1 _d)	(1 _d)
1s	2s	2p			3s			3p			4s	3d		

↑ print 8 314 C_r

a. VE correspond to the electrons after the previous noble gas.

b. VE are gained, lost or shared to form ionic or covalent bonds.

c. VE are the same for each element in a group on the periodic table.

1 wrong -5
2 wrong -15

N^{3-} Mg^{2+} Al^{3+} F^{-} Isoelectronic noble gas = Ne

Na⁺ vs. F⁻

Svs. Cl
A.R
P.T.

$$X(g) \longrightarrow X^+(g) + e^-$$

↳ if "an" $\neg$
b/c not Def of $\exists$

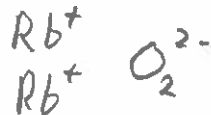
1 wrong $\rightarrow$
2 wrong $\rightarrow -1.5$

Si vs. O F vs. H S vs. C

Slides

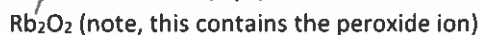
SHOW YOUR WORK

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Key

10. Provide the IUPAC names for the compounds below. (3 pt)



Chromium (II) Phosphide

Rubidium Peroxide

Magnesium Phosphate

11. ~ A3.151 Predict the chemical formula for the ionic compound formed by Fe^{3+} and the monatomic ion that N becomes. Writing the monatomic ion of N is an excellent partial credit move. (2 pt)



12. Identify the following elements as metals (M) or nonmetals (NM). (2 pt)

H

NM

C

NM

Pb

M

Li

M

13. Which of the following compound(s) exhibit(s) both covalent (molecular) and ionic bonding? Circle all correct choices. (2 pt)



Covalent bonds w/in polyatomic nitrate

14. What are the correct formulas for nickel (II) sulfate, calcium sulfide and sodium chloride? Write them below. (2.5 pt)

1 wrong
2 wrong

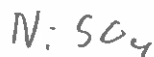


nickel (II) sulfate



calcium sulfide

sodium chloride



15. Identify the three bonds below as ionic bonds (IB), polar covalent bonds (PCB), or nonpolar covalent bonds (NCB). The same bond type might be used multiple times. (2 pt)

NCB C-H

Ca-O IB

O-O NCB

16. For the following elements: Ne, Br, Co, I, & K group them with the correct group name. Note, All elements will be used, meaning one spot will be filled with two symbols. (2.5 pt)

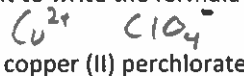
Transition Metals: Co

Alkali Metals: K

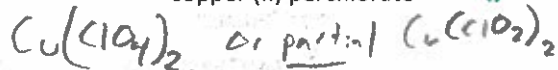
Halogens: Br, I

Noble/Inert Gases: Ne

17. Provide the correct chemical formula for the below name(s) or the correct IUPAC name (using the rules learned in class) for the given formula(s). Note, if you don't know what perchlorate is, you can use ClO_2^- for partial credit to write the formula instead. (3 pt)



Diphosphorus tetrachloride



Sulfur Hexafluoride

18. For hybridizing orbitals, 4 atomic orbitals in, 4 hybrid orbitals out. Fill in the blanks for a generic case or for the formation of sp^2 orbitals. (1 pt)



19. The image here depicts a sigma (σ), pi (π), or delta (δ) bond? Would this be a bonding or antibonding interaction in molecular orbital theory? The two dots signify electrons, orbitals, or nuclei. Circle your answer in each sentence. (2 pt)



2 wrong

π

Key

20. How many valence electrons do the elements below have? (2 pt)

Hydrogen

Carbon

Wrong (-)

21. Draw the Lewis symbols for the elements below using dots to show the required electrons. (1 pt)

• B •

Boron

“F”

Fluorine

22. Draw the Lewis structure of C_2H_2 . Identifying the total # of valence electrons is a good move. (2 pt)

$$2 \times 4 + 2 \times 1 = 10$$

2.5pt



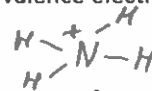
move. (2 pt) (1)
if $H - \bar{c} - \bar{c} - H$

a. The non-H atoms above follow the octet rule. Briefly define the octet rule, or circle "I used it correctly" below. (1 pt)

I used it correctly.

non-H atoms want $8e^-$
around them

23. Draw the Lewis structure for NH_4^+ . Identifying the total # of valence electrons is a good move. (3 pt)

$$5 + 4 \times 1 - (41) = 8$$


if $\text{H}-\text{N}(\text{H})_3$ (3)

a. Identify the formal charge for all atoms that have non-zero formal charges. (1 pt)

for N: $F.C. = \frac{\text{Valence } e}{5} - \frac{\text{Lines}}{4} - \text{Dots} = +1$

24. A Lewis structure is provided. Identify its electron-pair (electron domain) and molecular geometries and hybridization. Note the geometries *may* be the same. (2.5 pt)

1 wrong (-1)
2 (-1.5)

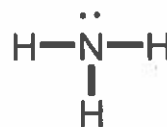
Electron-pair geometry

Molecular geometry

Hybridization

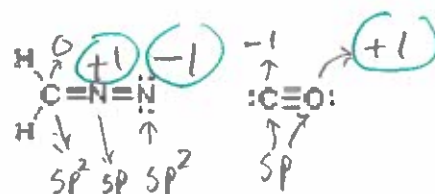
Fe tra hedral

Trigonal Pyramidal

 sp^3 

25. Calculate the formal charge on each non-hydrogen, non-carbon atom in the two given structures. (2 pt)

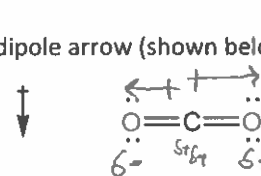
1 wrong (-1)



a. For any of the non-H atoms above, identify the hybridization. All identifications will be graded. (1 pt)

26. The Lewis structure for CO₂ is provided below. Draw both bond dipoles ^v a dipole arrow (shown below), or δ⁺ & δ⁻. Then identify if CO₂ is a **polar** or **nonpolar** molecule. Circle it. (3 pt)

1.5 pt.



a. ~~What is the strongest~~ intermolecular force present in methane (shown here)? Circle it. (1 pt)

Van der Waals Forces

Dipole-Dipole Forces

Hydrogen-Bonding.



b. Is $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$ or CH_3OCH_3 more soluble in methane? Briefly explain your answer. (1 pt)

Like Dissolves Like, weaker IMF's are more akin to those

c. Are all atoms in the above compound in the same 2D plane? Yes or No. (0.5 pt)

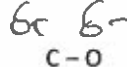
in CH_4

SHOW YOUR WORK

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Key

1. For each bond below, identify which atom has the δ^- or δ^+ charge. (1 pt)



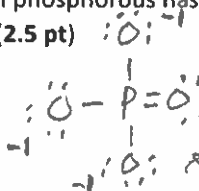
2. Briefly define the word **nomenclature**. Hint: IUPAC nomenclature. (2 pt)

→ a system for naming

3. Draw the Lewis structure of PO_4^{3-} where the central phosphorous has a formal charge = zero. Identifying the total number of valence electrons is a good move. (2.5 pt)

$$5 + 6 \times 4 - (-3) = 32$$

P O_4



if all P-O, but right

7.5

- a. Does the central phosphorous above have a formal charge = zero? **Yes** or no. Circle it. (0.5 pt)
- b. Do all atoms in your structure above, follow or are below the octet rule? **Yes** or **no**. Circle the correct bolded term. You must have a Lewis structure to be able to answer this question. (1 pt)
- c. Which of the elements in the above formula **cannot** be hypervalent (have an expanded octet)? Circle all that are correct below. (1 pt)

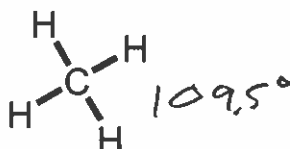
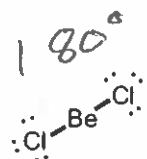
O

P

Neither can be hypervalent

period 2 elements must follow Octet rule

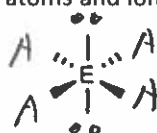
27. For the species below, identify the bond angles for each. (2 pt)



28. Identify how many atoms and lone pairs are present around a central atom (E) with the electron pair geometry of **square planar**. Drawing out atoms and lone pairs around E is a smart move. (2 pt)

Atoms

4



Lone Pairs

2

- a. Something with this geometry would have what **hybridization** and what **bond angle(s)**? (1 pt)

hybridization

sp^3d^2

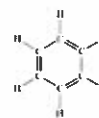
bond angle(s)

90°

29. In benzene, shown here, identify how many **sigma (σ)** and **pi (π)** bonds are present. (2 pt)

12

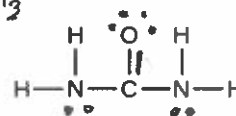
3



30. Why is the H-N-H angle in NH_3 smaller than the H-C-H bond angle in CH_4 ? (1 pt)

b/c lone pair "pushes" H's closer in NH_3

31. On the given structure for urea, add lone pairs, double bonds, or triple bonds, to make all atoms have a full octet and a formal charge of zero. (2 pt)



C w/ lone e⁻ (-2)

When needed/expected, **SHOW YOUR WORK**

Exam III Nov 13th 2025 Chem 1127

Name (Full First then Last or -1 pts):

Key

Page 1-4: Q's Page 5-6: Equations & Periodic Table

Points: ____/80₈₄

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When needed/expected, **SHOW YOUR WORK**

1. A species that is a base Accepts protons. While a species that is oxidized Loses electrons. Correctly fill in the blanks with the below words. A word may be used multiple times. Note if you would rather use a different word, you may, but the words below will work. (4 pt)

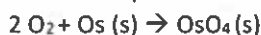
Hydroxide(s)	Proton(s)	Electron(s)	Neutron(s)	Hydronium(s)
Loses	Gains	Transmutes	Capitalizes	

Alternatively - "Creates" Hydroxide, if loses hydroxides -1.5

2. Clearly identify 3 strong acids and 2 strong bases in the space below. Names or formula are fine. (2 pt)

Slides

3. What mass of oxygen is required when 25.0 g of osmium reacts completely according to the balanced chemical equation below? (2 pt)



$$\frac{25 \text{ g Os}}{1} \cdot \frac{1 \text{ mol Os}}{190.2 \text{ g Os}} \cdot \frac{2 \text{ mol O}_2}{1 \text{ mol Os}} \cdot \frac{32 \text{ g O}_2}{1 \text{ mol O}_2} = 8.41 \text{ g O}_2$$

- a. Balanced chemical equations are in terms of relative amounts of: moles, molarities, molecules, or grams. Circle all reasonable bolded terms. (1.5 pts)

- b. The above problem requires you to determine molar masses (MM). Clearly identify one of the compounds and its correct MM. (0.5 pt)



$$16 \times 2 = \frac{32 \text{ g}}{\text{mol}}$$

- c. What phase of matter (s, l, g, or aq) is O_2 commonly observed as? (1 pt)

$$\text{OsO}_4$$

$$\text{MM} = 190.2 + 4 \times 16$$

$$= 254.2 \text{ g/mol}$$

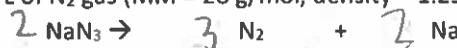
- d. 27 g of OsO_4 were produced from this reaction. Calculate the percent yield. Doing so requires you to calculate the actual yield or theoretical yield. Circle the correct term & calculate the percent. (2 pt)

$$\frac{25 \text{ g Os}}{1} \cdot \frac{1 \text{ mol Os}}{190.2 \text{ g Os}} \cdot \frac{1 \text{ mol OsO}_4}{1 \text{ mol Os}} \cdot \frac{254.2 \text{ g OsO}_4}{1 \text{ mol OsO}_4} = 33.4 \text{ g OsO}_4$$

33.4 g OsO₄ would be theoretically produced

$$\% \text{ yield} = \frac{27}{33.4} \times 100\% = 80.8\%$$

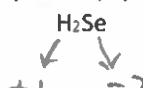
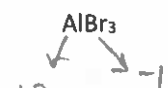
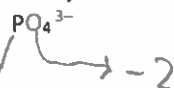
4. How many moles of NaN_3 are required to produce 48 L of N_2 gas (MM = 28 g/mol, density = 1.25 g/L) according to the chemical reaction: (2.5 pt)



$$\frac{48 \text{ L N}_2}{1} \cdot \frac{1.25 \text{ g N}_2}{1 \text{ L N}_2} \cdot \frac{1 \text{ mol N}_2}{28 \text{ g N}_2} \cdot \frac{2 \text{ mol NaN}_3}{3 \text{ mol N}_2} = 1.46 \text{ mol NaN}_3$$

- a. If the above equation is unbalanced, balance it, or circle already balanced. (1 pt)

5. Clearly identify the oxidation number (state) for each element below. Note, if the same element is given twice, it may have different oxidation numbers in different species. (5 pt)



- a. Oxidation numbers are like _____. Circle the phrase below that goes in the blank. (0.5 pt)

Electronegativity

Ionization Energy

Ionic Charge

Energy Transfer

- b. One of the above species is an ionic compound. Write its cation and anion with the correct ionic charge for each. (1 pt)



-5 for every one wrong unless all wrong -5

+5

b/c

$$+5 + 4(-2) = -3 \text{ charge for each}$$

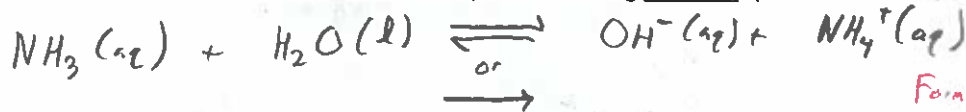
Key

When needed/expected, **SHOW YOUR WORK**

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When needed/expected, **SHOW YOUR WORK**

6. Aqueous ammonia (NH_3) is a weak base that reacts with liquid water to produce aqueous H_3O^+ or OH^- . Write the balanced reaction w/ phases. Note be sure to identify the remaining aqueous species that is also formed. (3 pt)



Forming NH_4^+ + H_2O (aq)
- no charges

7. What is wrong with your chemical experiment if you get a 111% yield? Briefly discuss. (2 pt)

"Y%" = if just "cant" find

Product is not pure,

8. List 6 of the memorized polyatomic formulas with the correct charge for each. (1 pt)

SO_4^{2-} , NO_3^- , CO_3^{2-} , PO_4^{3-} , H_2PO_4^- , HPO_4^{2-}

9. Is $\text{H}_3\text{PO}_4(\text{aq}) + \text{Ca}(\text{OH})_2(\text{aq}) \rightarrow \text{H}_2\text{O}(\text{l}) + \text{Ca}_3(\text{PO}_4)_2(\text{aq})$. Is this a redox, precipitation, or acid-base reaction? Circle the correct one and briefly explain your answer. (2.5 pt)

proton (H^+) transfer occurs

no expl. -1.5

- a. Solid gold (Au) dissolves upon reaction with $\text{HCl}(\text{aq})$, forming hydrogen gas and $\text{Au}(\text{l})$ or $\text{Au}^{3+}(\text{aq})$. Is this a redox, precipitation, or acid-base reaction? Circle the correct bolded term in each, and briefly explain your choice. (2 pt)

if Acid-base
HCl an acid, -1.5

Gold changes its oxidation state to dissolve

10. Cidal™ is a solution of 70% ethanol (volume by volume, v/v) in water. The company claims this is the optimal concentration of ethanol to kill bacteria. If 100 mL of solution are present, match the underlined terms to identify what the solution, solvent, and solute are. (2.5 pt)

Cidal Ethanol Water

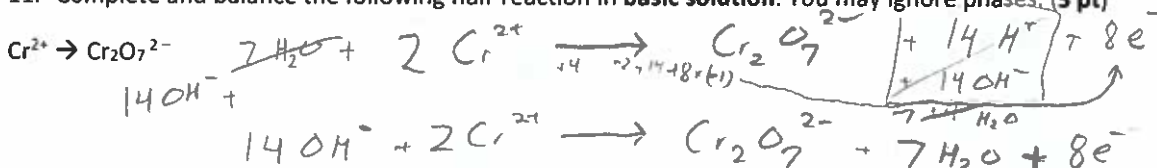
no expl. & Au^{3+} not circled -1.5

- a. What does the term aqueous signify for a solution? Briefly define it. (1 pt)

Species are dissolved in water

even only this, Full pt.

11. Complete and balance the following half-reaction in basic solution. You may ignore phases. (3 pt)



- a. Basic solution means which of the following may be present in your half-reaction: H^+ , OH^- , or H_2O . Circle all that may show up in a basic solution. (1 pt)

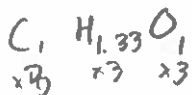
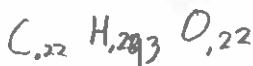
12. Complete and balance the following half-reaction. You may ignore phases. (2 pt)



- a. Identify whether this is a reduction or oxidation half-reaction. You must have a reasonable attempt in the main part to be able to earn this point. (1 pt)

if only this way, -1.5 $\rightarrow \text{e}^-$ gained.

13. In a balanced chemical equation or half-equation, Atoms & Net Charge must be balanced. Fill in the blanks with the correct terms/phrases. (1 pt)



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$$MM H_2O = 18$$

$$MM CO_2 = 44$$

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When needed/expected, **SHOW YOUR WORK**

14. A 6.49 g sample of an unknown compound composed of C, H, & O was burned in a combustion train. 9.74 g CO_2 and 2.64 g H_2O were formed. Determine the empirical formula of the unknown compound. (3 pt)

$$Mass Sample = Mass C + Mass H + Mass O$$

$$6.49 = 2.64g C + 0.293g H$$

$$Mass O = 6.49 - 2.64 - 0.293 = 3.557g$$

$$9.74g CO_2 \cdot \frac{1 mol CO_2}{44g} \cdot \frac{1 mol C}{1 mol CO_2} = 0.22 mol C = 2.64g$$

$$2.64g H_2O \cdot \frac{1 mol H_2O}{18g} \cdot \frac{2 mol H}{1 mol H_2O} = 0.293 mol H = 0.293g$$

- a. A necessary step in solving the main problem above is determining how many grams of hydrogen (H) are in 2.64 g H_2O . Determine it in the space below. (2 pt)

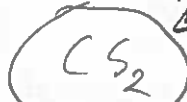
$$MM = 18g \rightarrow 0.222 mol O$$

Divide them all by 0.22

- b. A picture of a combustion analysis train is shown. A gas, CO_2 , H_2O , O_2 , or N_2 , enters from the right or left of the machine. Circle the correct bolded word in each. (1 pt)



15. Determine the empirical formula of a compound that is 15.8% carbon and 84.2% sulfur by mass. (2 pt)



$$15.8g C \div 12$$

$$84.2g S \div 32.1$$

$$\frac{C}{1.317} \quad \frac{S}{2.62} \quad \frac{1.317}{1.317} \quad \frac{2.62}{1.317}$$

- a. When working with percents, it is easiest to assume you have 100g g as a total mass. identify the number that should go here. Getting the main part earns you these pt. Hint: it is also the normal boiling point of water. (1 pt)
- b. Which of the following is (are) correct as an empirical formula(s)? Circle all that will be a proper empirical formula. (1 pt)



16. A compound has an empirical formula of CH_2O , and its molar mass is known to be 60 g/mol. What is its molecular formula? (3 pt)

$$MM \text{ of } EF = 12 + 2 + 16$$

$$60 \div 30 = 2$$

No work, (-2)



17. The end point in a titration of a 75 mL sample of aqueous HCl was reached by addition of 30 mL of 0.25 M $Ca(OH)_2$ titrant. The unbalanced titration reaction is below, Balance it. (1 pt)



- a. How many moles of $Ca(OH)_2$ are present in 30 mL of the 0.25 M $Ca(OH)_2$ titrant? (2 pt)

$$\frac{0.03L}{1} \cdot \frac{0.25 mol Ca(OH)_2}{1 L} = 0.0075 mol Ca(OH)_2$$

- b. What is the molarity (M) of HCl? Identify the two units that make up molarity for partial credit (3 pt)

$$\frac{0.0075 mol Ca(OH)_2}{1} \cdot \frac{2 mol HCl}{1 mol Ca(OH)_2} = 0.015 mol HCl \rightarrow \frac{0.015 mol HCl}{0.075 L \text{ of solution}} = 0.2 M HCl$$

- c. Circle the bolded term that is in the long thin glass buret for the above titration: H_2O , HCl (aq), $Ca(OH)_2$ (aq). Also draw in the liquid's level, assuming the given liquid was before any $Ca(OH)_2$ was added, when the end point is reached and label it c. (1 pt)



When needed/expected, **SHOW YOUR WORK**

Key

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When needed/expected, **SHOW YOUR WORK**

18. What is the concentration if you dilute 60 mL of 1.64 M NaI to a total volume of 0.75 L by adding pure solvent? Note, the units of volume. (2.5 pt)

$$C_i V_i = C_f V_f$$

$$C_f = \frac{C_i V_i}{V_f} = \frac{1.64 \cdot 60}{750} = 0.13M$$

- a. Identify which volume above corresponds to the more **dilute** solution in NaI. (1 pt)
- b. What amount of pure water was necessary for this dilution? (1 pt)

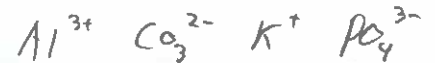
$$750 - 60 = 690 \text{ mL} = 0.69L$$

19. Balance the following equation. (1 pt) $\text{Al}_2(\text{CO}_3)_3(\text{aq}) + 2 \text{K}_3\text{PO}_4(\text{aq}) \rightarrow 2 \text{AlPO}_4(\text{s}) + 3 \text{K}_2\text{CO}_3(\text{aq})$

- a. Identify which compound above is (are) **soluble**. all identifications are graded. (2 pt)

Only ONE missing - (-1.5) ~~all~~ that are labeled (aq) but $\text{Al}_2(\text{CO}_3)_3$ is technically insoluble if you look at the table

- b. Identify all the ions that are present with correct ionic charges. (2 pt)



20. Calculate the theoretical yield in moles NH_3 from the complete reaction of 59.8 grams N_2 and 30.2 g H_2 according to the balanced chemical equation: $\text{N}_2(\text{g}) + 3 \text{H}_2(\text{g}) \rightarrow 2 \text{NH}_3(\text{g})$. (2.5 pt)

$$\frac{59.8 \text{ g } \text{N}_2}{1} \cdot \frac{1 \text{ mol } \text{N}_2}{28 \text{ g } \text{N}_2} \cdot \frac{2 \text{ mol } \text{NH}_3}{1 \text{ mol } \text{N}_2} = 4.27 \text{ mol } \text{NH}_3$$

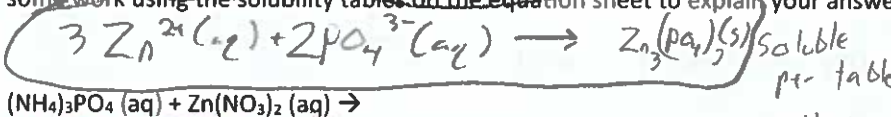
$$\frac{30.2 \text{ g } \text{H}_2}{1} \cdot \frac{1 \text{ mol } \text{H}_2}{2 \text{ g } \text{H}_2} \cdot \frac{2 \text{ mol } \text{NH}_3}{3 \text{ mol } \text{H}_2} = 10.1 \text{ mol } \text{NH}_3$$

- a. The above reaction relates to the Haber Bosch process. Describe why this process or reaction is important. (1 pt)

It makes Fertilizer (NH_3) plant food.

Uses a lot of Humanity's energy. Mildly inefficient 10-20% yield.

21. Complete the balanced **net ionic** equation for the precipitation reaction below. If no reaction occurs, write NR after the reaction arrow. Be sure to include all proper phases for all species in the reaction. You should show some work using the solubility tables on the equation sheet to explain your answer. (4 pt)



Phases 1 pt

soluble per table
insoluble per table

22. Which of the following ions would form a precipitate with Pb^{2+} ions in aqueous solution? Circle all correct ions. (1 pt)



Always soluble

23. A particular polypeptide is determined to have 2 cobalt atoms in one molecule and was determined to be 4.34% cobalt by mass. What is the molar mass of the peptide? (2 pt)

$$\frac{4.34 \text{ g Co}}{100 \text{ g}}$$

$$\text{assume } 100 \text{ g molecule} \rightarrow \frac{4.34 \text{ g Co}}{1} \cdot \frac{1 \text{ mol Co}}{58.9 \text{ g Co}} \cdot \frac{1 \text{ mol molecule}}{2 \text{ mol Co}} = 0.0368 \text{ mol compound}$$

$$\text{MM} = \frac{100 \text{ g}}{0.0368 \text{ mol}} = 2714 \text{ g/mol}$$

ID'ing 58.9 x 2 → 0.5 pt